

OFDM: Complete Mathematics of the One-Shot Receivers in JunaCore

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Abstract—This note derives, end to end, the mathematics of the one-shot Standard OFDM receiver that anchors the JunaCore family: the cyclic-prefix argument that reduces a convolutive channel to one complex gain per subcarrier, the per-pilot least-squares channel estimate, the declared piecewise-linear interpolation operator in active-carrier rank coordinates with endpoint hold, zero-forcing equalization, soft demapping under a pooled pilot-residual noise scale, and normalized min-sum LDPC decoding. Each stage is classified the way the receiver-hierarchy analysis demands: the least-squares and zero-forcing steps are exact conditional minimizers; the interpolation operator and the pooled log-likelihood-ratio scale are declared engineering choices whose implicit assumptions are stated, not hidden. Part II develops the full theory of the Partial-FFT + FEC receiver: the exact decomposition of the FFT statistic into M temporal views, the induced branch leakage kernels and noise statistics, the band-tied pilot-trained ridge combiner with its uniqueness proposition and pilot-count identifiability contract, the residual per-carrier equalization that reuses the Part I operator, and the validity-gated standard-FFT fallback. Both receivers share the same demapper and decoder; neither feeds the decoder back, and the failure boundary of one-shot reception marks the entry point of the JUNA refinement receivers. Part III states what lies beyond that boundary: the exact maximum a posteriori problem for joint channel and codeword inference under intercarrier coupling, its relaxation into the seven-family cost function that couples ICI reduction, decision-domain combining, and code structure, the provenance of each family (likelihood, discriminative training, prior, or surrogate), and the ideal variational form whose sum-product fixed points are Bethe stationary points. The joint criterion is a hybrid generative–discriminative design objective, not a posterior; stating that distinction exactly is what allows the refinement receivers to be classified as exact specializations, conditional minimizers, or approximations.

Index Terms—OFDM, channel estimation, pilot interpolation, zero-forcing equalization, LLR, LDPC, min–sum decoding, partial-FFT, intercarrier interference, MAP estimation, Bethe free energy.

I. INTRODUCTION

THE JunaCore family has two one-shot receivers. The Standard OFDM baseline (Part I, Sections II–VII) is pilot least squares, piecewise-linear interpolation, per-carrier division, one LDPC decode. The Partial-FFT + FEC baseline (Part II, Sections VIII–XI) replaces the single FFT statistic by M temporal views and one exact pilot-trained ridge combiner per frequency band. In both, nothing flows backward — a decoding failure cannot repair the front end. Their value is twofold: they are cheap, and every stage admits a complete mathematical description with explicitly stated assumptions. Part III (Sections XII–XIV) then states the joint inference

Technical note, Acoustic Research Laboratory. Every equation is synchronized with the executable JunaCore implementation; deployed constants (LLR clip 20, scale floor 0.02, min–sum factor 0.8) cite the shipped defaults.

problem that the one-shot structure leaves unsolved — combining, ICI reduction, and LDPC decoding in one cost function — first as an exact MAP problem, then as the relaxed hybrid criterion the JUNA refinement receivers actually descend. This note supplies those descriptions, keeping each symbol synchronized with the executable implementation.

II. SIGNAL MODEL AND THE DIAGONAL REDUCTION

Consider one OFDM symbol with N subcarriers, cyclic prefix length N_{cp} , unitary DFT matrix $\mathbf{F}$ with entries $F_{nk} = N^{-1/2} e^{-j2\pi nk/N}$, pilot set $\mathcal{P}$, and data set $\mathcal{D}$. The transmitted frequency-domain vector $\mathbf{X} \in \mathbb{C}^N$ maps to the time-domain block $\mathbf{x} = \mathbf{F}^H \mathbf{X}$, prepended by its last N_{cp} samples. The channel is an L -tap linear filter $\mathbf{h} = (h_0, \dots, h_{L-1})$ with additive noise $\boldsymbol{\nu} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$.

If (i) the channel is time invariant over the block and (ii) $L - 1 \leq N_{\text{cp}}$, then CP removal converts linear convolution into circular convolution: the received block is $\mathbf{y} = \mathbf{H}_{\text{cir}} \mathbf{x} + \boldsymbol{\nu}$ with circulant $\mathbf{H}_{\text{cir}}$. Circulant matrices are diagonalized by the DFT, so

$$\mathbf{Y} = \mathbf{F} \mathbf{H}_{\text{cir}} \mathbf{F}^H \mathbf{X} + \mathbf{F} \boldsymbol{\nu} = \text{diag}(\mathbf{H}) \mathbf{X} + \tilde{\boldsymbol{\nu}}, \quad (1)$$

where the carrier gains are samples of the channel transfer function,

$$H_k = \sum_{l=0}^{L-1} h_l e^{-j2\pi kl/N}, \quad (2)$$

and $\tilde{\boldsymbol{\nu}} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$ because $\mathbf{F}$ is unitary. Every subsequent stage assumes the scalar-per-carrier model $Y_k = H_k X_k + \nu_k$. Both hypotheses are load bearing: hypothesis (ii) fails when delay spread exceeds the CP, and hypothesis (i) fails under residual Doppler, in which case energy from carrier q leaks into observation k and (1) acquires off-diagonal terms (Section XI).

III. PILOT LEAST SQUARES

On each pilot carrier $p \in \mathcal{P}$ the transmitted value P_p is known. The receiver estimates each pilot gain independently.

Proposition 1 (Per-pilot LS). *For the scalar model $Y_p = H_p P_p + \nu_p$ with $P_p \neq 0$, the least-squares problem $\min_{H \in \mathbb{C}} |Y_p - H P_p|^2$ has the unique minimizer*

$$\hat{H}_p = \frac{Y_p}{P_p}. \quad (3)$$

Proof. The objective is a positive-definite quadratic in H ; the Wirtinger stationarity condition $P_p^*(Y_p - H P_p) = 0$ gives (3). $\square$

The estimate is unbiased, $\mathbb{E}[\hat{H}_p] = H_p$, with variance $\text{Var}(\hat{H}_p) = \sigma^2/|P_p|^2$; for unit-modulus pilots this is exactly σ^2 . No smoothing across pilots is applied at this stage — each pilot contributes one noisy sample of (2).

IV. PIECEWISE-LINEAR INTERPOLATION

Data-carrier gains are produced by a fixed linear operator applied to the pilot estimates. The implementation works in *active-carrier rank* coordinates: let $r(k)$ enumerate the active carriers in order (DC and guard carriers excluded), and let $r_1 < \dots < r_Q$ be the pilot ranks. For a data carrier with rank r :

$$\hat{H}(r) = \begin{cases} \hat{H}_{r_1}, & r \leq r_1, \\ (1-t)\hat{H}_{r_i} + t\hat{H}_{r_{i+1}}, & r_i < r \leq r_{i+1}, \\ \hat{H}_{r_Q}, & r > r_Q, \end{cases} \quad (4)$$

with $t = (r - r_i)/(r_{i+1} - r_i)$. Equivalently $\hat{\mathbf{H}} = \mathbf{G}_{\text{lin}}\hat{\mathbf{H}}_{\mathcal{P}}$ for a row-stochastic matrix $\mathbf{G}_{\text{lin}}$ with at most two nonzero entries per row (one at the endpoints). Two properties deserve emphasis.

It is a declared prior, not a derived one. Nothing in the least-squares stage implies linear interpolation; (4) is the receiver's smoothness assumption about (2), equivalent to a linear-spline model of H between pilot ranks with zeroth-order hold outside them. Different interpolants define different receivers.

Its error is controlled by delay spread times pilot spacing. Treating H as a function of continuous rank, (2) gives the curvature bound $|\partial_k^2 H| \leq \sum_l |h_l| (2\pi l/N)^2$, and the standard linear-interpolation bound yields, for pilot spacing $\Delta = r_{i+1} - r_i$,

$$|H(r) - \hat{H}^{\text{interp}}(r)| \leq \frac{\Delta^2}{8} \sum_{l=0}^{L-1} |h_l| \left(\frac{2\pi l}{N}\right)^2 + \text{noise}, \quad (5)$$

where the noise term is the interpolated pilot-estimation error with variance at most σ^2 per carrier for unit-modulus pilots (a convex combination of two independent estimates has variance $((1-t)^2 + t^2)\sigma^2 \leq \sigma^2$). Long channels (large l) and sparse pilots (large Δ) therefore degrade the estimate quadratically, which is the classical reason pilot density must track delay spread. The endpoint hold replaces the quadratic term by a first-order one proportional to the distance beyond the outermost pilot.

V. ZERO-FORCING EQUALIZATION

Proposition 2 (Conditional LS symbol estimate). *Given $\hat{H}_k \neq 0$, the least-squares problem $\min_{x \in \mathbb{C}} |Y_k - \hat{H}_k x|^2$ has the unique minimizer*

$$\hat{s}_k = \frac{Y_k}{\hat{H}_k}. \quad (6)$$

The implementation applies (6) only when $|\hat{H}_k|$ exceeds machine epsilon and passes the raw carrier through otherwise. Substituting the true model, $\hat{s}_k = s_k (H_k/\hat{H}_k) + \nu_k/\hat{H}_k$: zero forcing is unbiased when the channel estimate is exact but

enhances noise on weak carriers, with post-equalization error variance

$$\text{Var}(\hat{s}_k - s_k) \approx \frac{\sigma^2}{|H_k|^2} + \underbrace{|s_k|^2 \frac{|H_k - \hat{H}_k|^2}{|H_k|^2}}_{\text{estimation error}}. \quad (7)$$

An MMSE equalizer $\hat{H}_k^* Y_k / (|\hat{H}_k|^2 + \sigma^2)$ would trade bias for variance; the standard receiver deliberately keeps the unbiased one-shot form.

VI. SOFT DEMAPPING WITH A POOLED PILOT-RESIDUAL SCALE

Data symbols are Gray-mapped QPSK, $s = \frac{1}{\sqrt{2}}((1-2b_I) + j(1-2b_Q))$, or BPSK with the real component only. If the post-equalization error were $\mathcal{CN}(0, \sigma_k^2)$ with known variance, the exact channel LLR of b_I (log ratio of $b_I=1$ to $b_I=0$) would be

$$L_I^{\text{exact}}(k) = -\frac{2\sqrt{2} \text{Re}(\hat{s}_k)}{\sigma_k^2}, \quad (8)$$

and correspondingly for b_Q with the imaginary part. The deployed receiver replaces the per-carrier variance by one pooled, self-calibrating scale estimated from the pilots after equalization:

$$\hat{\beta} = \max \left(\frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} |\hat{s}_p - P_p|^2, \beta_{\min} \right), \quad \beta_{\min} = 0.02, \quad (9)$$

$$L_I(k) = \text{clip}_{\pm 20} \left(-\frac{2 \text{Re}(\hat{s}_k)}{\hat{\beta}} \right), \\ L_Q(k) = \text{clip}_{\pm 20} \left(-\frac{2 \text{Im}(\hat{s}_k)}{\hat{\beta}} \right). \quad (10)$$

Three deliberate approximations separate (10) from (8). First, *pooling*: $\hat{\beta}$ is one number, while (7) says the true error variance varies per carrier as $\sigma^2/|H_k|^2$; a matched demapper would weight strong carriers more. Second, *self-calibration*: $\hat{\beta}$ measures the total pilot residual — thermal noise, interpolation error, and any model mismatch — so when the diagonal model degrades, every LLR automatically weakens, which is a robustness feature rather than an accident. Third, *clipping and flooring*: the clip at ± 20 bounds decoder input dynamic range, and the floor $\beta_{\min}$ prevents unbounded confidence when few pilots produce a near-zero residual by chance. The constant absorbed between 2 and $2\sqrt{2}$ is calibrated into $\hat{\beta}$ itself.

VII. NORMALIZED MIN-SUM LDPC DECODING

The coded bits satisfy $\mathbf{H}_{\text{ldpc}} \mathbf{b} = \mathbf{0}$ over $\mathbb{F}_2$. Decoding is iterative message passing on the Tanner graph with variable messages $m_{v \rightarrow c}$ and check messages $m_{c \rightarrow v}$, initialized from the channel LLRs L_v of Section VI. The deployed check update is the *normalized min-sum* rule with factor $\alpha = 0.8$:

$$m_{c \rightarrow v} = \alpha \left(\prod_{v' \in \mathcal{N}(c) \setminus v} \text{sign } m_{v' \rightarrow c} \right) \min_{v' \in \mathcal{N}(c) \setminus v} |m_{v' \rightarrow c}|, \quad (11)$$

$$m_{v \rightarrow c} = L_v + \sum_{c' \in \mathcal{N}(v) \setminus c} m_{c' \rightarrow v},$$

$$L_v^{\text{post}} = L_v + \sum_{c' \in \mathcal{N}(v)} m_{c' \rightarrow v}, \quad (12)$$

with all messages clipped to ± 20 . Hard decisions $\hat{b}_v = \mathbb{1}[L_v^{\text{post}} < 0]$ are checked against the syndrome each iteration; decoding stops at $\mathbf{H}_{\text{ldpc}} \hat{\mathbf{b}} = \mathbf{0}$ or at the iteration cap. Rule (11) approximates the exact sum-product update $2 \operatorname{artanh}[\prod \tanh(m/2)]$; the min-sum envelope overestimates check reliability, and the factor $\alpha < 1$ compensates on average. The receiver reports the candidate with its validity flag and syndrome weight; in the standard receiver this decode is terminal — its posteriors influence nothing upstream.

PART II: PARTIAL-FFT + FEC

VIII. PARTIAL-FFT OBSERVATION VECTORS

The partial-FFT front end [1] replaces the single FFT statistic by M *temporal views*. The implementation partitions the CP-stripped block of N samples into M contiguous segments $\mathcal{I}_p = \{a_p, \dots, a_p + L_p - 1\}$ of near-equal length ($M = 4$ deployed), zero-pads each segment to length N , and takes a full FFT:

$$y_{p,k} = \frac{1}{\sqrt{N}} \sum_{n \in \mathcal{I}_p} r[n] e^{-j2\pi kn/N}, \quad \mathbf{y}_k = (y_{1,k}, \dots, y_{M,k})^\top. \quad (13)$$

Because the segment indicators form a partition of unity,

$$\sum_{p=1}^M y_{p,k} = Y_k \quad \text{for every } k: \quad (14)$$

the decomposition is an exact identity, not an approximation — the partial FFT invents no energy, it splits the full FFT statistic into views whose *relative* amplitudes and phases carry the within-block time-variation information that the sum destroys.

Two consequences follow directly. First, the branch noise statistics: since the segment windows have disjoint support and the DFT normalization is unitary, the stacked noise $\boldsymbol{\nu}_k = (\nu_{1,k}, \dots, \nu_{M,k})^\top$ satisfies

$$\operatorname{Cov}(\boldsymbol{\nu}_k) = \sigma^2 \operatorname{diag}(L_1/N, \dots, L_M/N), \quad (15)$$

independent across branches. Second, the branch response to a physical path: for a path of delay τ_r , residual Doppler ν_r , and gain α_r , substituting the modulated carrier into (13) and summing the geometric series gives the branch leakage kernel

$$y_{p,k} = \sum_r \alpha_r \sum_q e^{-j2\pi f_q \tau_r} \psi_{kq}^{(p)}(\tau_r, \nu_r) X_q + \nu_{p,k}, \quad (16)$$

with $\psi_{kq}^{(p)}(\tau, \nu) = D_{L_p, N}(q - k + \nu N T_s; a_p)$, where $D_{L_p, N}(\xi; a)$ is the windowed Dirichlet kernel $\frac{e^{j\pi\xi(2a+L-1)/N} \sin(\pi L \xi/N)}{N \sin(\pi \xi/N)}$. The segment offset a_p enters only through the phase factor: under residual Doppler, *each view sees the same leakage magnitude but a segment-dependent phase rotation* $e^{j\pi\xi(2a_p+L_p-1)/N}$ — effectively sampling the channel's time variation at M epochs. When the channel is block-constant ($\nu_r = 0$, time-invariant), the kernels collapse so that every view carries the same information and combining reduces to the scalar model of Part I.

IX. BAND-TIED PILOT RIDGE COMBINING

The receiver forms one linear combiner per frequency band. The active carriers are split by rank into B contiguous bands ($B = 16$ deployed; $B = 4$ in the frame profile), and for k in band b the equalized symbol is

$$\hat{s}_k = \mathbf{w}_b^H \mathbf{y}_k. \quad (17)$$

The weights are trained only on the band's pilots $\mathcal{P}_b$ with a ridge penalty:

$$\hat{\mathbf{w}}_b = \arg \min_{\mathbf{w} \in \mathbb{C}^M} \sum_{p \in \mathcal{P}_b} |\mathbf{w}^H \mathbf{y}_p - P_p|^2 + \delta \|\mathbf{w}\|_2^2, \quad (18)$$

with deployed ridge $\delta = 10^{-3}$.

Proposition 3 (Exact band combiner). *For $\delta > 0$ the objective in (18) is a positive-definite quadratic with unique global minimizer*

$$\hat{\mathbf{w}}_b = \left(\sum_{p \in \mathcal{P}_b} \mathbf{y}_p \mathbf{y}_p^H + \delta \mathbf{I} \right)^{-1} \sum_{p \in \mathcal{P}_b} \mathbf{y}_p P_p. \quad (19)$$

Proof. The Hessian $2(\sum_p \mathbf{y}_p \mathbf{y}_p^H + \delta \mathbf{I})$ is positive definite for $\delta > 0$; Wirtinger stationarity gives the normal equations (19). $\square$

The implementation accumulates exactly these $M \times M$ normal equations per band and solves them by Gaussian elimination with partial pivoting; the solve is an *exact conditional minimizer* in the sense of the receiver hierarchy, contract-tested against an independent dense solution. Identifiability is enforced as a validity precondition rather than hoped for: a configuration is accepted only if every band contains at least M pilot rows, so the M unknowns of (19) are never fit from fewer equations than parameters (the ridge would otherwise silently determine the deficit directions).

Two further deployed details complete the pipeline. First, the combined output still passes through the *residual* per-carrier equalizer of Part I — pilot division (3) followed by rank-linear interpolation (4) applied to $\hat{\mathbf{s}}$ — because $\mathbf{w}_b$ is constant across a band and cannot express per-carrier gain and phase; the Part I operator restores exactly that diagonal freedom on top of the band-tied view combination. Second, a *validity-gated fallback*: if the Partial-FFT candidate fails LDPC validity, the receiver also evaluates the one-view standard candidate of Part I and uses it only when valid, so view combination can never hide a conventional decode that would have succeeded. Demapping and decoding are then identical to Sections VI–VII, with the pooled scale $\hat{\beta}$ computed from the same post-equalization pilot residuals.

X. WHAT TIME DIVERSITY BUYS, AND WHAT IT CANNOT

Equation (16) explains the mechanism: under within-block time variation the M views are differently phased copies of the same leakage structure, and a per-band M -tap combiner can re-align them — first-order intercarrier-interference mitigation without ever modeling the coupling explicitly. The cost side is equally explicit. The combiner is trained on pilots alone, so it optimizes pilot prediction, not data prediction; band tying assumes the optimal view weighting varies slowly in

frequency; the leakage *within* each segment is untouched; and the noise weighting (15) is implicit in the normal equations rather than whitened. Most fundamentally the receiver remains one-shot: when the pilot-trained combiner is wrong for the data region, the decoder discovers it and can do nothing — repairing exactly that is where the JUNA receivers begin.

XI. THE FAILURE BOUNDARY OF ONE-SHOT RECEPTION

Every Part I stage conditions on the diagonal model (1). Under residual time scaling or within-block channel variation the true model gains intercarrier coupling, $Y_k = \sum_q C_{kq} s_q + \nu_k$: pilot division (3) then estimates not H_p but a symbol-dependent mixture, interpolation (4) smooths estimates of the wrong quantity, zero forcing (6) cannot remove coupling by scalar division, and the pooled scale (9) correctly senses the damage but can only weaken every LLR uniformly. The receiver degrades exactly as its assumptions predict. Part II repairs the first-order damage: temporal views recover within-block variation information and the band combiner re-aligns it. What neither receiver can do is react — both are one-shot, so the decoder's discovery of a failure arrives too late to help. Feeding decoder posteriors back into combiner training and ultimately modeling the coupling itself is precisely the program of the JUNA receivers; the taxonomy of those refinements and their optimization machinery is treated in the companion notes.

PART III: THE JOINT OBJECTIVE

XII. THE EXACT MAP PROBLEM

Under residual Doppler the honest observation model is coupled. Stacking the M partial-FFT views of Section VIII, carrier k obeys

$$\mathbf{y}_k = \sum_{q \in \mathcal{N}_K(k)} \mathbf{c}_{kq} x_q(\mathbf{b}) + \nu_k, \quad \nu_k \sim \mathcal{CN}(\mathbf{0}, \Sigma_k), \quad (20)$$

where $\mathcal{N}_K(k) = \{q : |q - k| \leq K\}$ is the coupling neighborhood, $\mathbf{c}_{kq} \in \mathbb{C}^M$ is the branch response of (16) (structured by delay-Doppler atoms or free), and the data symbols $x_q(\mathbf{b})$ are generated by the coded bits $\mathbf{b} \in \{0, 1\}^n$ satisfying $\mathbf{H}_{\text{ldpc}} \mathbf{b} = \mathbf{0}$. With a prior $p(\mathbf{C})$ on the channel and a uniform prior over codewords, the *exact* joint maximum a posteriori problem is

$$\min_{\mathbf{C}, \mathbf{b}} \sum_k \left\| \mathbf{y}_k - \sum_{q \in \mathcal{N}_K(k)} \mathbf{c}_{kq} x_q(\mathbf{b}) \right\|_{\Sigma_k^{-1}}^2 - \log p(\mathbf{C}) - \log \mathbb{P}\{\mathbf{H}_{\text{ldpc}} \mathbf{b} = \mathbf{0}\}. \quad (21)$$

Three facts about (21) shape everything downstream. First, it contains *no combiner*: $\mathbf{W}$ is a decision statistic, not a latent variable of the forward model, so a true MAP treatment has nothing to say about it. Second, it is intractable: the codeword set is exponential, and message passing computes approximate marginals, not the joint maximizer. Third, without structure it is unidentifiable: an unconstrained per-carrier $\mathbf{c}_{kq}$ offers $M(2K+1)$ unknowns against M observations at each carrier, so the observation term can be driven to zero for *any* bit hypothesis and then carries no information about $\mathbf{b}$. Exact MAP is therefore the reference point, not the algorithm.

XIII. THE JOINT DESIGN CRITERION

The deployed family relaxes (21) in three moves. Bits become logits $\mathbf{z} \in \mathbb{R}^n$ with bipolar means $\xi_i = \tanh(z_i/2)$ and soft QPSK symbols $s_q(\mathbf{z})$; the code indicator becomes the differentiable parity surrogate over check neighborhoods $\{\mathcal{N}_j\}$,

$$\mathcal{R}_{\text{par}}(\mathbf{z}) = \sum_{j=1}^m \left(1 - \prod_{i \in \mathcal{N}_j} \xi_i(\mathbf{z}) \right)^2, \quad (22)$$

which vanishes exactly when every check is satisfied at hard bits; and the decision statistic $\hat{s}_k = \mathbf{w}_{b(k)}^H \mathbf{y}_k$ is added with its own calibration losses. The result is the seven-family joint criterion

$$\begin{aligned} \mathcal{J}(\mathbf{C}, \mathbf{W}, \mathbf{z}) = & \lambda_d \sum_k \left\| \mathbf{y}_k - \sum_{q \in \mathcal{N}_K(k)} \mathbf{c}_{kq} s_q(\mathbf{z}) \right\|_{\Sigma_k^{-1}}^2 \\ & + \eta \sum_{p \in \mathcal{P}} \left| \mathbf{w}_{b(p)}^H \mathbf{y}_p - P_p \right|^2 \\ & + \rho \sum_{d \in \mathcal{D}} \left| \mathbf{w}_{b(d)}^H \mathbf{y}_d - s_d(\mathbf{z}) \right|^2 \\ & + \gamma_C \|\mathbf{C}\|_F^2 + \gamma_W \|\mathbf{W}\|_F^2 \\ & + \gamma_S \sum_b \|\mathbf{w}_{b+1} - \mathbf{w}_b\|_2^2 + \lambda_p \mathcal{R}_{\text{par}}(\mathbf{z}). \end{aligned} \quad (23)$$

The first family is the *ICI-reduction cost*: it explains each observation vector through the coupled model, transferring energy from unexplained residual into modeled coupling; minimized over a structured $\mathbf{C}$ it is the only family with a direct Gaussian-likelihood parent in (21). The pilot and tie families train the decision statistic — discriminative losses with no likelihood parent. The three regularizers are conditioning priors, and the parity family is a declared smooth surrogate of the code indicator. Table I records this provenance. Consequently (23) is a *hybrid generative-discriminative design criterion, not a posterior*: calling it MAP would claim a probability model that does not exist for $\mathbf{W}$, and the companion hierarchy paper builds the receiver classification on exactly this distinction.

Identifiability survives only through structure. Tying $\mathbf{c}_{kq}$ and $\mathbf{w}$ across each of the B frequency bands restores an overdetermined conditional channel fit whenever a band holds more than $2K+1$ carriers; pilots pin the scale-and-phase gauge that could otherwise migrate freely between $\mathbf{C}$, $\mathbf{W}$, and $\mathbf{z}$; and the parity family couples symbols globally, which is what lets code structure add training equations beyond the physical pilots. These are counting and gauge arguments, not a joint identifiability theorem; the precise local condition is a bilinear Jacobian rank statement given in the companion paper.

XIV. IDEAL VARIATIONAL FORM AND MESSAGE PASSING

Direct gradient optimization of (22) is poorly conditioned for high-degree checks, so the deployed receivers let BP carry the code side. The idealized joint problem replaces the point estimate of $\mathbf{b}$ by beliefs $q(\mathbf{b})$ with symbol means $\mathbb{E}_q[X_k]$:

$$\begin{aligned} \mathcal{V}(\mathbf{C}, q) = & \mathbb{E}_q[-\log p(\mathbf{Y} | \mathbf{X}, \mathbf{C})] - \log p(\mathbf{C}) \\ & + \mathbb{E}_q[-\log P_{\text{code}}(\mathbf{b})] - H_B(q), \end{aligned} \quad (24)$$

TABLE I
PROVENANCE OF THE SEVEN FAMILIES IN (23).

Family	Provenance
Observation (ICI reduction)	Gaussian likelihood of (20)
Pilot calibration	discriminative (decision statistic)
Combiner tie	discriminative (decision statistic)
$\mathbf{C}$, $\mathbf{W}$ regularizers	conditioning priors
Frequency smoothness	conditioning prior
Parity surrogate	smooth relaxation of the code indicator

TABLE II
MATHEMATICAL STATUS OF EACH STAGE.

Stage	Equation	Status
Pilot LS	(3)	exact minimizer
Interpolation	(4)	declared operator
Zero forcing	(6)	exact conditional minimizer
Soft demap	(10)	declared approx. of (8)
LDPC decode	(11)–(12)	approximate fixed point
Branch split	(14)	exact identity
Band combiner	(19)	exact minimizer
Residual equalize	(3)+(4)	declared operator (reused)
Standard fallback	—	validity-gated control rule

where P_{code} is the uniform-over-codewords prior and H_B the Bethe entropy approximation; the channel evidence enters only through the first term, so nothing is counted twice. Interior fixed points of ideal sum–product BP are stationary points of (24) at fixed channel evidence [2]; the deployed normalized min–sum rule of (11) is an approximate projection, not an exact block minimizer. The alternation this suggests — with beliefs fixed, the $\mathbf{C}$ and $\mathbf{W}$ subproblems of (23) are convex quadratics solvable exactly; with the front end fixed, BP updates the beliefs — is the blueprint of every JUNA refinement receiver, and the one-shot receivers of Parts I–II are its degenerate cases: $\rho = \lambda_d = 0$ with pilot-only $\mathbf{W}$ recovers (18), and $M = 1$, $K = 0$ with a scalar channel fit recovers Part I. Which restriction of (23) each refinement receiver descends, and with what optimization machinery, is the subject of the companion taxonomy and hierarchy notes.

XV. SUMMARY

Table II classifies every stage of both receivers. Three are exact conditional minimizers with propositions and machine-checkable normal equations; one is an exact algebraic identity; the interpolation operator and the pooled error scale are declared engineering choices whose implicit assumptions are stated and bounded; the decoder is a fixed-point iteration that approximates sum–product decoding with a documented normalization; and the fallback is a validity-gated control rule. This is the complete mathematical content of both pipelines: nothing else is computed.

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